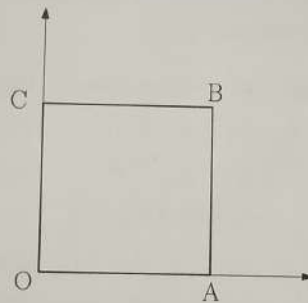


**EE596H (Optimization Techniques): Half Semester
Examination**

Total Questions: 5, Total Marks: 50, Time: 180 Mins

1. Let OABC be a square plate of side 4 meters. Taking O as the origin, let (x, y) be any point on the plate. Let the temperature at the point (x, y) be given by

$$T(x, y) = 2xy - 2x^2 - 2y^2.$$



It is desired to find the hottest point on the plate. Formulate the above as an optimization problem and solve the same to get the desired point.

[8]

2. Use the KKT theorem to find the value of β for which $x_1 = 1, x_2 = 2$ is optimal to the problem

$$\begin{aligned} &\text{maximize} && 2x_1 + \beta x_2 \\ &\text{s.t.} && x_1^2 + x_2^2 \leq 9 \\ &&& x_1^2 - x_2 \leq 2 \end{aligned}$$

[10]

3. Consider the following problem:

$$\begin{array}{ll}\min & x_1^2 + x_1x_2 + x_2^2 \\ \text{s.t.} & x_1^2 + x_2^2 = 1\end{array}$$

- (a) Is the above objective function a convex function?
 (b) Is the feasible set a convex set?
 (c) Does strong duality hold? Explain and elaborate.

3

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Total: 12

4. Write the dual program of the following primal problem (no need to solve the dual, simply write the dual problem):

$$\begin{array}{ll}\text{minimize} & x_1^2 + x_2^2 - 10x_1 - 10x_2 \\ \text{subject to} & x_1 + x_2 \leq 12 \\ & x_1 - x_2 \leq 6 \\ & x \geq 0, x_2 \geq 0\end{array}$$

It is claimed that the optimal value of the dual problem is -10. Do you agree with the claim? Justify your answer using the duality theorem.

8

5. Consider the following linear fractional programming problem:

$$\begin{array}{ll}\min & \frac{3x_1 + 4x_2 + 10}{10x_1 + 20x_2 + 10} \\ \text{s.t.} & x_1 + 3x_2 \leq 50 \\ & 3x_1 + 2x_2 \leq 80 \\ & x_1, x_2 \geq 0\end{array}$$

- (a) Transform the above problem to an equivalent linear programming problem.
 (b) Derive the KKT conditions for the same and find the solution. If it is not possible, then explain.

6

6

Total: 12